

# CALCULUS FOR ENGINEERS

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The vector function  $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$  has a **derivative (is differentiable)** at  $t$  if  $f, g$ , and  $h$  have derivatives at  $t$ . The derivative is the vector function  $\mathbf{r}'(t) = \frac{d\mathbf{r}}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\mathbf{r}(t+\Delta t) - \mathbf{r}(t)}{\Delta t} = \frac{df}{dt}\mathbf{i} + \frac{dg}{dt}\mathbf{j} + \frac{dh}{dt}\mathbf{k}$

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The tangent line to the curve at a point  $(f(t_0), g(t_0), h(t_0))$  is defined to be the line through the point parallel to  $\mathbf{r}'(t_0)$ .



# Derivative of a vector

## Definition

*If  $\mathbf{r}$  is the position vector of a particle moving along a smooth curve in space, then*

- 1. Velocity is the derivative of position:  $\mathbf{v} = \frac{d\mathbf{r}}{dt}$ .*
- 2. Speed is the magnitude of velocity:  $\text{Speed} = |\mathbf{v}|$ .*
- 3. Acceleration is the derivative of velocity:  $\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$ .*
- 4. The unit vector  $\mathbf{v}/|\mathbf{v}|$  is the direction of motion at time  $t$ .*

**Example** Find the velocity, speed, and acceleration of a particle whose motion in space is given by the position vector

$$\mathbf{r}(t) = 2 \cos t \mathbf{i} + 2 \sin t \mathbf{j} + 5 \cos^2 t \mathbf{k}.$$

# Derivative of a vector

## Differentiation Rules for Vector Functions

Let  $\mathbf{u}$  and  $\mathbf{v}$  be differentiable vector functions of  $t$ ,  $\mathbf{C}$  a constant vector,  $c$  any scalar, and  $f$  any differentiable scalar function.

1. *Constant Function Rule:*  $\frac{d}{dt}\mathbf{C} = \mathbf{0}$
2. *Scalar Multiple Rules:*  
 $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$   
 $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
3. *Sum Rule:*  $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
4. *Difference Rule:*  $\frac{d}{dt}[\mathbf{u}(t) - \mathbf{v}(t)] = \mathbf{u}'(t) - \mathbf{v}'(t)$
5. *Dot Product Rule:*  $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
6. *Cross Product Rule:*  $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
7. *Chain Rule:*  $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

# Integrals of Vector functions

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The **indefinite integral** of  $\mathbf{r}$  with respect to  $t$  is the set of all anti-derivatives of  $\mathbf{r}$ , denoted by  $\int \mathbf{r}(t)dt$ . If  $\mathbf{R}$  is any anti-derivatives of  $\mathbf{r}$ , then

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If the components of  $\mathbf{r}(t) = f(t)\mathbf{i} + g(t)\mathbf{j} + h(t)\mathbf{k}$  are integrable over  $[a, b]$ , then so is  $\mathbf{r}$ , and the **definite integral** of  $\mathbf{r}$  from  $a$  to  $b$  is

$$\int_a^b \mathbf{r}(t)dt = \left( \int_a^b f(t)dt \right) \mathbf{i} + \left( \int_a^b g(t)dt \right) \mathbf{j} + \left( \int_a^b h(t)dt \right) \mathbf{k}.$$

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**Example** Integrate  $\int_0^\pi ((\cos t)\mathbf{i} + \mathbf{j} - 2t\mathbf{k})dt$

# Integrals of Vector functions

**Example** Suppose we do not know the path of a hang glider, but only its acceleration vector  $\mathbf{a}(t) = -(3 \cos t)\mathbf{i} - (3 \sin t)\mathbf{j} + 2\mathbf{k}$ . We also know that initially (at time  $t = 0$ ) the glider departed from the point  $(4, 0, 0)$  with velocity  $\mathbf{v}(0) = 3\mathbf{j}$ . Find the glider's position as a function of  $t$ .

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*Solution*

$$\mathbf{r}(t) = (1 + 3 \cos t)\mathbf{i} + (3 \sin t)\mathbf{j} + t^2\mathbf{k}.$$



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## Definition

The **length** of a smooth curve  $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ ,  $a \leq t \leq b$ , that is traced exactly once as  $t$  increases from  $t = a$  to  $t = b$ , is

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt = \int_a^b |v| dt$$

**Example** A glider is soaring upward along the helix  $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$ . How long is the glider's path from  $t = 0$  to  $t = 2\pi$ ?

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$$L = \int_a^b |v| dt = 2\pi\sqrt{2}.$$

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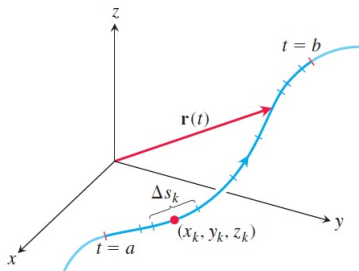
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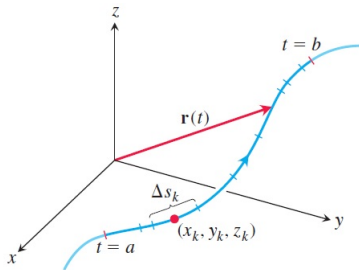
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## Definition

If  $f$  is defined on a curve  $C$  given parametrically by

$\mathbf{r}(t) = g(t)\mathbf{i} + h(t)\mathbf{j} + k(t)\mathbf{k}$ ,  $a \leq t \leq b$ , then the line integral of  $f$  over  $C$  is

$$\int_C f(x, y, z) ds = \int_a^b f(g(t), h(t), k(t)) |v(t)| dt$$

$$\frac{ds}{dt} = |v(t)| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}$$

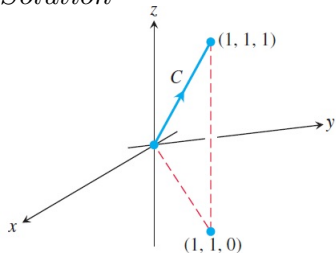


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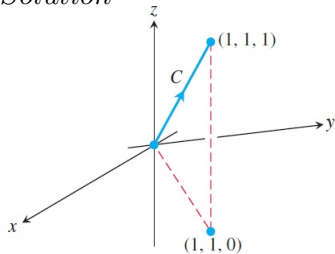
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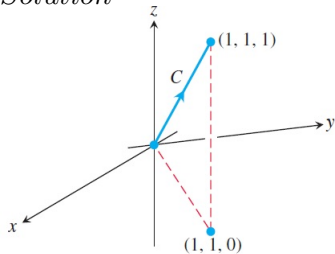
$$\text{So } |v(t)| = |i + j + k| = \sqrt{3}.$$

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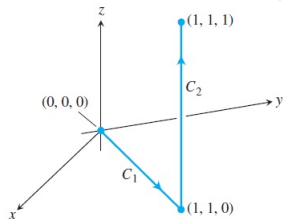
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**Note**  $\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds + \cdots + \int_{C_n} f ds.$

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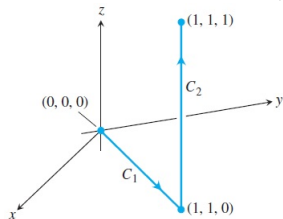
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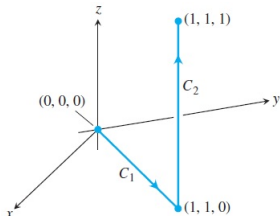
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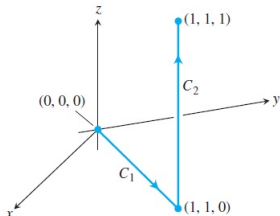
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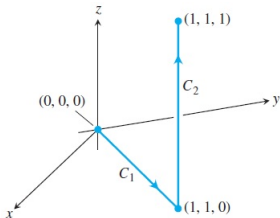
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*Solution*  $v(t) = \mathbf{r}'(t) = -\sin t \mathbf{i} + \cos t \mathbf{j} + \mathbf{k}$  and

$$|v(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1} = \sqrt{2}$$

The function along the path  $f(\mathbf{r}(t)) = f(\cos t, \sin t, t) = \sin 2t + \sqrt{t}$ .

$$\int_C f(x, y, z) ds = \int_0^\pi (\sin 2t + \sqrt{t}) \sqrt{2} dt = \frac{2\sqrt{2}}{3} \pi^{3/2} \simeq 5.25$$

# Mass and Moment calculations

In a coil springs or wires masses distributed along smooth curves in space. The density function  $\delta(x, y, z)$  representing mass per unit length. The curve  $C$  is parametrized by  $r(t) = x(t)i + y(t)j + z(t)k, a \leq t \leq b$ .

The arc length  $ds = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$

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**Moments of inertia about axes:**  $I_x = \int_C (y^2 + z^2) \delta ds, \quad I_y = \int_C (x^2 + z^2) \delta ds, \quad I_z = \int_C (x^2 + y^2) \delta ds, \quad I_L = \int_C r^2 \delta ds, \quad .$

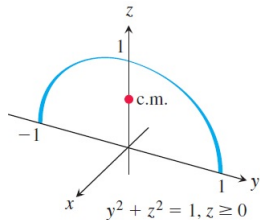
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**Example** A slender metal arch, denser at the bottom than top, lies along the semicircle  $y^2 + z^2 = 1, z \geq 0$ , in the  $yz$ -plane. Find the center of the arch's mass if the density at the point  $(x, y, z)$  on the arch is  $\delta(x, y, z) = 2 - z$ .

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*Solution*

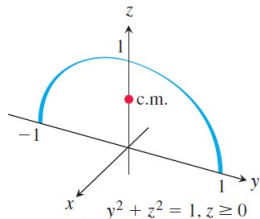




# Mass and Moment calculations

**Example** A slender metal arch, denser at the bottom than top, lies along the semicircle  $y^2 + z^2 = 1, z \geq 0$ , in the  $yz$ -plane. Find the center of the arch's mass if the density at the point  $(x, y, z)$  on the arch is  $\delta(x, y, z) = 2 - z$ .

*Solution*



Here  $\bar{x} = 0$  and  $\bar{y} = 0$  because the arch lies in the  $yz$ -plane.

$r(t) = \cos t \mathbf{j} + \sin t \mathbf{k}$ , and  $0 \leq t \leq \pi$ .

$$|v(t)| = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = 1,$$

so  $ds = |v|dt = dt$ .

$$M = \int_C \delta ds = \int_C (2 - z) ds = \int_0^\pi (2 - \sin t) dt = 2\pi - 2$$

$$M_{xy} = \int_C z \delta ds = \int_C z(2 - z) ds = \int_0^\pi (\sin t)(2 - \sin t) dt = \frac{8 - \pi}{2}.$$

$$\bar{z} = \frac{M_{xy}}{M} = \frac{8 - \pi}{4\pi - 4} \simeq 0.57$$

# Line Integrals

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**Example** A slender rod of constant density lies along the line segment  $\mathbf{r}(t) = t\mathbf{j} + (2 - 2t)\mathbf{k}$ ,  $0 \leq t \leq 1$ , in the  $yz$ -plane. Find the moments of inertia of the rod about the three coordinate axes.

# Line Integrals of Vector Fields

## Definition

Let  $F$  be a vector field with continuous components defined along a smooth curve  $C$  parametrized by  $r(t)$ ,  $a \leq t \leq b$ . Then the **line integral of  $F$  along  $C$**  is

$$\int_C F \cdot T ds = \int_C \left( F \cdot \frac{dr}{ds} \right) ds = \int_C F \cdot dr.$$

## Evaluating the Line Integral of $F = Mi + Nj + Pk$ along

$C : r(t) = g(t)i + h(t)j + k(t)k$

1. Express the vector field  $F$  in terms of the parametrized curve  $C$  as  $F(r(t))$  by substituting the components  $x = g(t)$ ,  $y = h(t)$ ,  $z = k(t)$  of  $r$  into the scalar components  $M(x, y, z)$ ,  $N(x, y, z)$ ,  $P(x, y, z)$  of  $F$ .
2. Find the derivative (velocity) vector  $dr/dt$ .
3. Evaluate the line integral with respect to the parameter  $t$ ,  $a \leq t \leq b$ , to obtain

## Line Integrals with respect to $dx, dy$ , or $dz$

$$\int_C F \cdot dr = \int_a^b F(f(t)) \cdot \frac{dr}{dt} dt$$

**Example** Evaluate  $\int_C F \cdot dr$ , where  $F(x, y, z) = zi + xyj - y^2k$  along the curve  $C$  given by  $r(t) = t^2i + tj + \sqrt{t}k$ ,  $0 \leq t \leq 1$ .

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$$\int_C F \cdot dr = \int_0^1 F(r(t)) \frac{dr}{dt} dt = \int_0^1 (2t^{3/2} + t^3 - \frac{1}{2}t^{3/2}) dt = \frac{17}{20}.$$

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$$\int_C P(x, y, z) dz = \int_a^b P(g(t), h(t), k(t)) k'(t) dt$$

# Line Integrals

**Example** Evaluate the line integral  $\int_C -ydx + zdy + 2xdz$ , where  $C$  is the helix  $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}$ ,  $0 \leq t \leq 2\pi$ .

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**Example** Find the work done by the force field

$\mathbf{F} = (y - x^2)\mathbf{i} + (z - y^2)\mathbf{j} + (x - z^2)\mathbf{k}$  along the curve  $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k}$ ,  $0 \leq t \leq 1$ , from  $(0, 0, 0)$  to  $(1, 1, 1)$ .

# Flow Integrals and Circulation for Velocity Fields

*Solution*

$$\mathbf{F} \cdot \frac{d\mathbf{r}}{dt} = [(t^3 - t^4)\mathbf{j} + (t - t^6)\mathbf{k}] \cdot (\mathbf{i} + 2t\mathbf{j} + 3t^2\mathbf{k}) = 2t^4 - 2t^5 + 3t^3 - 3t^8.$$

$$\text{Work} = \int_0^1 (2t^4 - 2t^5 + 3t^3 - 3t^8) dt = \frac{29}{60}.$$

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# Flow Integrals and Circulation for Velocity Fields

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## Definition

If  $\mathbf{r}(t)$  parametrizes a smooth curve  $C$  in the domain of a continuous velocity field  $\mathbf{F}$ , the **flow** along the curve  $A = \mathbf{r}(a)$  to  $B = \mathbf{r}(b)$  is

$$\text{Flow} = \int_C \mathbf{F} \cdot \mathbf{T} ds.$$

The integral is called a **flow integral**. If the curve starts and ends at the same point, so that  $A = B$ , the flow is called the **circulation** around the curve.

**Example** A fluid's velocity field is  $\mathbf{F} = x\mathbf{i} + z\mathbf{j} + y\mathbf{k}$ . Find the flow along the helix  $\mathbf{r}(t) = (\cos t)\mathbf{i} + (\sin t)\mathbf{j} + t\mathbf{k}, 0 \leq t \leq \pi/2$ .

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*Solution*  $\mathbf{F} = (\cos t - \sin t)\mathbf{i} + (\cos t)\mathbf{j}$ , and  $\frac{d\mathbf{r}}{dt} = (-\sin t)\mathbf{i} + (\cos t)\mathbf{j}$ .

$$\text{Circulation} = \int_0^{2\pi} \mathbf{F} \cdot \frac{d\mathbf{r}}{dt} dt = \int_0^{2\pi} (1 - \sin t \cos t) dt = 2\pi.$$

## Flux across a simple closed plane curve

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## Definition

A curve is said to be **simple** if it does not cross itself. When a curve starts and ends at the same points, it is a **closed curve** or **loop**.



Simple,  
not closed



Simple,  
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Not simple,  
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# Flux across a simple closed plane curve

## Definition

*If  $C$  is a smooth simple closed curve in the domain of a continuous vector field  $\mathbf{F} = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$  in the plane, and if  $\mathbf{n}$  is the outward-pointing unit normal vector on  $C$ , the **flux** of  $\mathbf{F}$  across  $C$  is*

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Flux of  $\mathbf{F} = M\mathbf{i} + N\mathbf{j}$  across  $C = \oint_C Mdy - Ndx$ .

The integral can be evaluated from any smooth parametrization

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**Example** Find the flux of  $\mathbf{F} = (x - y)\mathbf{i} + x\mathbf{j}$  across the circle  $x^2 + y^2 = 1$  in the  $xy$ -plane.

## Flux across a simple closed plane curve

*Solution* The parametrization  $\mathbf{r}(t) = \cos t \mathbf{i} + \sin t \mathbf{j}$ ,  $0 \leq t \leq 2\pi$ , traces the circle counterclockwise exactly once. So

$$M = x - y = \cos t - \sin t, \quad N = x = \cos t \quad dy = \cos t dt, \quad dx = -\sin t dt, \\ \text{Flux} = \oint_C M dy - N dx = \int_0^{2\pi} \cos^2 t dt = \pi.$$

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The flux of  $\mathbf{F}$  across the circle is  $\pi$ . Since the answer is positive, the net flow across the curve is outward. A net inward flow would have given a negative flux.

# Path Independence, Conservative Fields, and Potential functions

## Definition

Let  $\mathbf{F}$  be a vector field defined on an open region  $D$  in space, and suppose that for any two points  $A$  and  $B$  in  $D$  the line integral  $\int_C \mathbf{F} \cdot d\mathbf{r}$  along a path  $C$  from  $A$  to  $B$  in  $D$  is the same over all paths from  $A$  to  $B$ . Then the integral  $\int_C \mathbf{F} \cdot d\mathbf{r}$  is **path independent in  $D$**  and the field  $\mathbf{F}$  is **conservative on  $D$** .

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## Definition

If  $\mathbf{F}$  is a vector field defined on  $D$  and  $\mathbf{F} = \nabla f$  for some scalar function  $f$  on  $D$ , then  $f$  is called a **potential function for  $\mathbf{F}$** .

# Path Independence, Conservative Fields, and Potential functions

## Definition

A domain  $D$  is called **connected**, if any two points in  $D$  can be joined by a smooth curve that lies in the region.

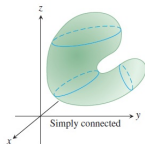
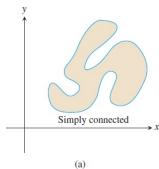
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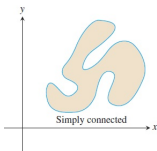


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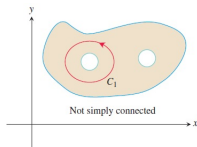
## Definition

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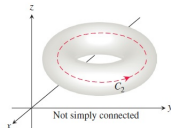
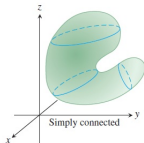
A domain is said to be **Simply connected**, if every loop in  $D$  can be contracted to a point in  $D$  without ever leaving  $D$ .



(a)



(c)



# Line Integrals in Conservative Fields

## Theorem (Fundamental Theorem of line integrals)

*Let  $C$  be a smooth curve joining the point  $A$  to the point  $B$  in the plane or in space and parametrized by  $\mathbf{r}(t)$ . Let  $f$  be a differentiable function with a continuous gradient vector  $\mathbf{F} = \nabla f$  on a domain  $D$  containing  $C$ . Then  $\int_C \mathbf{F} \cdot d\mathbf{r} = f(B) - f(A)$ .*

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**Example** Suppose the force field  $\mathbf{F} = \nabla f$  is the gradient of the function  $f(x, y, z) = -\frac{1}{x^2+y^2+z^2}$ . Find the work done by  $\mathbf{F}$  in moving an object along a smooth curve  $C$  joining  $(1, 0, 0)$  to  $(0, 0, 2)$  that does not pass through the origin.

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*Solution*  $\int_C \mathbf{F} \cdot d\mathbf{r} = f(0, 0, 2) - f(1, 0, 0) = 3/4$ .

# Line Integrals in Conservative Fields

## Theorem (Conservative fields are Gradient fields)

*Let  $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$  be a vector field whose components are continuous throughout an open connected region  $D$  in space. Then  $\mathbf{F}$  is conservative if and only if  $\mathbf{F}$  is a gradient field  $\nabla f$  for a differentiable function  $f$ .*

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# Component Test for Conservative fields

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*Let  $\mathbf{F} = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$  be a field on an open simply connected domain whose component functions have continuous first partial derivatives. Then,  $\mathbf{F}$  is conservative if and only if*

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}.$$

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**Example** Show that  $\mathbf{F} = (e^x \cos y + yz)\mathbf{i} + (xz - e^x \sin y)\mathbf{j} + (xy + z)\mathbf{k}$  is conservative over its natural domain and find a potential function for it.

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*Solution* Here  $M = e^x \cos y + yz$ ,  $N = xz - e^x \sin y$ ,  $P = xy + z$ .

$$\frac{\partial P}{\partial y} = x = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = y = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = -e^x \sin y + z = \frac{\partial M}{\partial y}.$$

The partial derivatives are continuous, so  $\mathbf{F}$  is conservative.  $\nabla f = \mathbf{F}$

$$\frac{\partial f}{\partial x} = e^x \cos y + yz, \quad \frac{\partial f}{\partial y} = xz - e^x \sin y, \quad \frac{\partial f}{\partial z} = xy + z.$$

$$f(x, y, z) = e^x \cos y + xyz + g(y, z).$$